

Recursive Algorithms

(5)

$A(n)$

{ if $(n > 1)$

return $A(n-1)$

}

constant time.

anchor
Cond.

$$T(n) = 1 + T(n-1) \quad \text{--- (1)} \quad T(n) = 1; n=0;$$

Back Substitution

$$T(n-1) = 1 + T(n-1-1) \\ = 1 + T(n-2) \quad \text{--- (2)}$$

$$T(n-2) = 1 + T(n-2-1) \\ = 1 + T(n-3) \quad \text{--- (3)}$$

Sub (2) in (1)

$$T(n) = 1 + T(n-1) \\ = 1 + 1 + T(n-2) \\ = 2 + T(n-2) \quad \text{--- (4)}$$

Sub (3) in (4)

$$= 1 + 1 + 1 + T(n-3) \\ = 3 + T(n-3)$$

$$\vdots \\ = k + T(n-k)$$

$$= (n-1) + T(n-(n-1))$$

$$= n-1 + T(1)$$

$$= n-1 + T(1)$$

$$= n-1 + 1$$

$$= n$$

$$= O(n)$$

$$n-k=1 \\ \boxed{k=n-1}$$

Ex (2)

$$T(n) = n + T(n-1); n > 1 \\ = 1; n = 1$$

$$T(n-1) = n-1 + T(n-1-1)$$

$$= (n-1) + T(n-2)$$

$$T(n-2) = (n-2) + T(n-3)$$

$$\text{(2) in (1)} \quad T(n) = n + T(n-1) \\ = n + (n-1) + T(n-2)$$

$$= n + (n-1) + (n-2) + T(n-3)$$

$$= n + (n-1) + (n-2) + \dots + (n-k) + T(n-k+1) \quad \text{--- (4)}$$

$$\begin{aligned}
 n - (k+1) &= 1 \\
 n - k - 1 &= 1 \\
 \boxed{k = n-2}
 \end{aligned}$$

Sub (k) in (4)

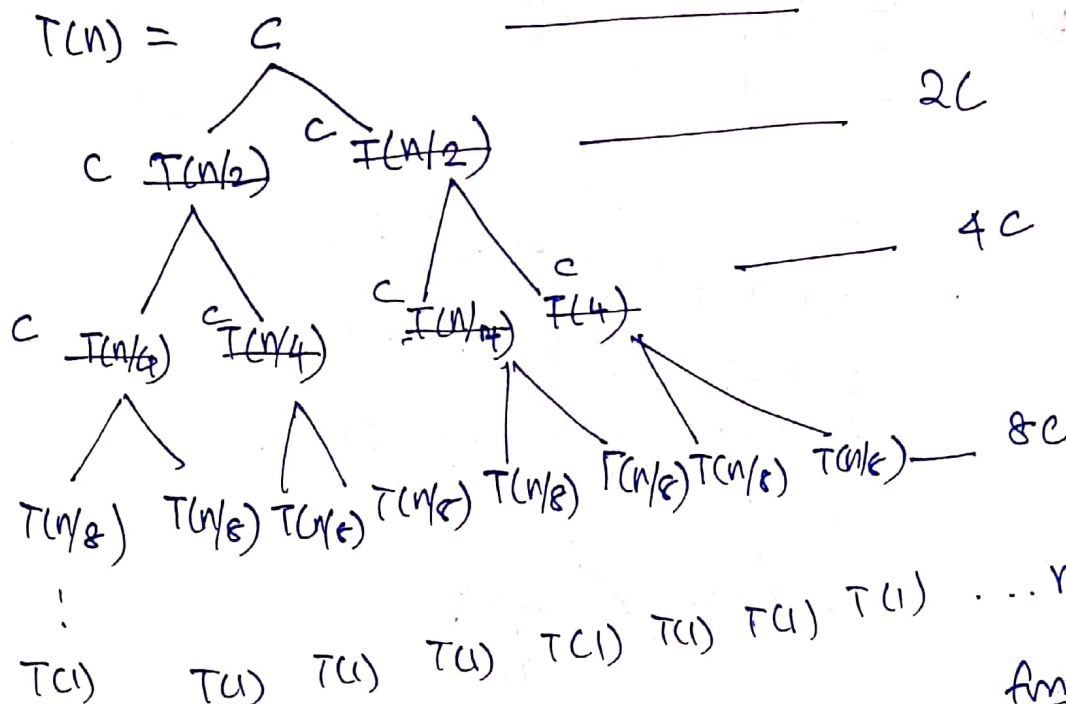
$$\begin{aligned}
 T(n) &= n + (n-1) + (n-2) + \dots + (n - (n-2)) + T(n - (n-2+1)) \\
 &= n + (n-1) + (n-2) + \dots + 2 + T(n - n + 2 - 1) \\
 &= n + (n-1) + (n-2) + \dots + 2 + 1 \\
 &= \frac{n(n+1)}{2} \\
 &= \underline{\underline{O(n^2)}}
 \end{aligned}$$

Recursion Tree Method

constant amount of work

$$\begin{aligned}
 T(n) &= 2T(n/2) + C \quad ; n > 1 \\
 &= C \quad ; n = 1
 \end{aligned}$$

How much work done



$$T(1) = T(n/n)$$

$$\begin{aligned}
 \text{Total work} &= C + 2C + 4C + \dots + nC \\
 &= C(1 + 2 + 4 + \dots + n) \\
 &= C(1 + 2^1 + 2^2 + \dots + 2^k) \\
 &= \cancel{C \frac{1 - 2^{k+1}}{2-1}} = C \frac{(2^{k+1} - 1)}{(2-1)} = C(2^{k+1} - 1) \\
 &= \underline{\underline{O(n)}}
 \end{aligned}$$

Formula

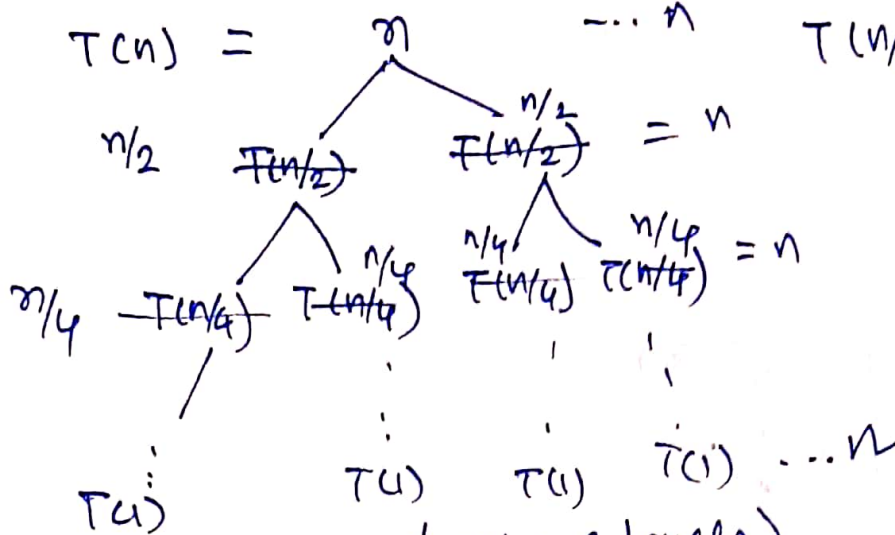
$$\frac{a(r-1)}{(r-1)}$$

Eg $T(n) = 2T(n/2) + n ; n > 1$

$= 1$

$T(n/2) = 2T(n/4) + n/2$

$T(n/4) = 2T(n/8) + n/4$



$\Rightarrow n \times (\text{number of levels})$

$\Rightarrow n \times (\log n + 1)$

$\Rightarrow n \log n + n$

$\Rightarrow O(n \log n)$

how many levels?

$n/2^0 - n/2^1 - n/2^2 \dots n/2^k$

$(k+1) \text{ level } n = 2^k$

$\underline{\underline{(\log n + 1)}}$ $\underline{\underline{k = \log n}}$

Master's Theorem

$T(n) = aT(n/b) + \Theta(n^k \log^p n)$

$a \geq 1, b > 1, k \geq 0$ and p is real no.

1) if $a > b^k$; then $T(n) = \Theta(n^{\log_b a})$

2) if $a = b^k$

a) if $p > -1$ then $T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$

b) if $p = -1$ then $T(n) = \Theta(n^{\log_b a} \log \log n)$

c) if $p < -1$ then $T(n) = \Theta(n^{\log_b a})$

3) if $a < b^k$

a) if $p \geq 0$ then $T(n) = \Theta(n^{k \log^p n})$

b) if $p < 0$; then $T(n) = \Theta(n^k)$

[Note $(\log n)^2 \neq \log^2 n$
 $= \log \log n$]

Eg $T(n) = 3T(n/2) + n^2$

$a=3 \quad b=2 \quad k=2 \quad p=0$

$3 < 2^2 \quad a < b^k$

$3 < 4$

case iii) holds

$\therefore P=0$

Cond'n a holds (3a)

$\therefore T(n) = \Theta(n^2 \log^0 n)$
 $= \Theta(n^2)$

② $T(n) = 4T(n/2) + n^2$

$a=4 \quad b=2 \quad k=2 \quad p=0$

$4 = 2^2 \quad a = b^k$

Case ii holds

$\therefore P=0$ (2a) holds

$T(n) = \Theta(n^{\log_2 4} \log^0 n)$
 $= \underline{\underline{\Theta(n^2 \log n)}}$

③ $T(n) = T(n/2) + n^2$

$a=1 \quad b=2 \quad k=2 \quad p=0$

$1 < 2^2 \quad a < b^k$

$\therefore P=0$ Case iii) a

$T(n) = \Theta(n^2 \log^0 n)$
 $= \Theta(n^2)$

④ $T(n) = 2^n T(n/2) + n^n$
 $\therefore a$ is not in the form,
 we cannot apply Master's Theorem

⑤ $T(n) = 16T(n/4) + n$
 $a=16 \quad b=4 \quad k=1 \quad p=0$
 $16 > 4^1$

Case 1 $T(n) = \Theta(n \log_b a)$
 $= \Theta(n \log_4 16)$
 $= \Theta(n^2)$

⑥ $T(n) = 2T(n/2) + n \log n$

$a=2 \quad b=2 \quad k=1 \quad p=1$

$a < b^k$

$2 < 2^1$

Case ii) a $T(n) = \Theta(n^{\log_b a} \log^{b+1} n)$
 $= \Theta(n^{\log_2 2} \log^2 n)$
 $= \Theta(n^1 \log^2 n)$
 $= \Theta(n \log^2 n)$

⑦ $T(n) = 2T(n/2) + n/\log n$
 $= 2T(n/2) + n \log^{-1} n$

$a=2 \quad b=2 \quad k=1 \quad p=-1$

$2 = 2^1$

Case ii) b holds $T(n) = \Theta(n^{\log_b a} \log \log n)$
 $= \Theta(n^{\log_2 2} \log \log n)$
 $= \Theta(n \log \log n)$

$$8) T(n) = 2T(n/4) + n^{0.51}$$

$$a=2 \quad b=4 \quad k=0.51 \quad p=0$$

$$2 < 4^{0.51}$$

Case iii a holds $T(n) = \Theta(n^k \log^p n)$

$$= \Theta(n^{0.51} \log^0 n)$$

$$= \Theta(n^{0.51})$$

$$9) T(n) = 0.5T(n/2) + 1/n$$

$$a=0.5 \quad \therefore a < 1$$

we cannot apply Master's Theorem

$$10) T(n) = 6T(n/3) + n^2 \log n$$

$$a=6 \quad b=3 \quad k=2 \quad p=1$$

$$6 < 3^2 \quad \underline{\text{Case 3a}} \text{ holds}$$

$$T(n) = \Theta(n^k \log^p n)$$

$$= \Theta(n^2 \log^1 n)$$

$$= \Theta(n^2 \log n)$$

$$11) T(n) = 64T(n/8) - n^2 \log n$$

$\therefore$ minus we cannot apply Master's Theorem.

$$12) T(n) = 7T(n/3) + n^2$$

$$a=7 \quad b=3 \quad k=2 \quad p=0$$

$$a > b^k$$

$$7 < 3^2$$

Case iii a holds

$$T(n) = \Theta(n^k \log^p n)$$

$$= \Theta(n^2 \log^0 n)$$

$$= \Theta(n^2)$$

$$13) T(n) = 4T(n/2) + \log n$$

$$a=4 \quad b=2 \quad k=0 \quad p=1$$

$$a > b^k$$

$$4 > 2^0$$

Case 1

$$T(n) = \Theta(n^{\log_b a})$$

$$= \Theta(n^{\log_2 4})$$

$$= \Theta(n^2)$$

$$(14) T(n) = \sqrt{2} T(n/2) + \log n$$

$$a = \sqrt{2} \quad b = 2 \quad k = 0 \quad p = 1$$

$$\sqrt{2} > 2^0$$

$$\text{Case 1} \quad T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 \sqrt{2}}) = \underline{\underline{\Theta(\sqrt{n})}}$$

$$(15) T(n) = 2T(n/2) + \sqrt{n}$$

$$a = 2 \quad b = 2 \quad k = \frac{1}{2} \quad p = 0$$

$$2 > 2^{\frac{1}{2}} \quad \text{Case 1} \quad T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 2}) = \underline{\underline{\Theta(n)}}$$

$$(16) T(n) = 3T(n/2) + n$$

$$a = 3 \quad b = 2 \quad k = 1 \quad p = 0$$

$$3 > 2^1 \quad T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 3})$$

$$(17) T(n) = 3T(n/3) + \sqrt{n}$$

$$a = 3 \quad b = 3 \quad k = \frac{1}{2} \quad p = 0$$

$$3 > 3^{\frac{1}{2}} \quad \text{Case 1} \quad T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_3 3}) = \underline{\underline{\Theta(n)}}$$

$$(18) T(n) = 4T(n/2) + Cn$$

$$a = 4 \quad b = 2 \quad k = 1 \quad p = 0$$

$$a > b^k \quad \text{Case 1} \quad T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 4}) = \underline{\underline{\Theta(n^2)}}$$

$$(19) T(n) = 3T(n/4) + n \log n$$

$$a = 3 \quad b = 4 \quad k = 1 \quad p = 1$$

$$3 < 4^1 \quad \text{Case 3a} \quad T(n) = \Theta(n^{k \log_b a}) = \Theta(n^1 \log^1 n) = \underline{\underline{\Theta(n \log n)}}$$